

Fqtrader — Complete PC Usage Guide

Scope: everything you do on your PC to run, monitor, and improve the freqtrade trading system in `ftbot/`. This is the operational companion to the strategy research in `ftbot/RESEARCH_returns.md` (the *why*) — this document is the *how*.

Cloud vs PC. Claude Code cloud sessions cannot reach Bybit or run freqtrade (no exchange/market-data access, no TA-Lib). Everything in §3–§7 runs on **your PC**. Cloud sessions are for analysis, code, and planning only.

Golden rule. The live paper bot runs **ClaudeBreakout**. Do **not** change its parameters or logic during the paper-trading window — that invalidates the experiment. All tuning happens on the *research* variants in backtests (§6), and is promoted only after the go-live gates pass (§8). See `CLAUDE.md`.

1. What you are running

Layer	What	Where
Trading engine	freqtrade ClaudeBreakout, 4h Donchian breakout, dry-run	your PC, 24/7
Phone control	freqtrade's native Telegram bot	your phone
Research / R&D	backtests, hyperopt, analysis	PC (run) + Claude cloud (interpret)

Strategy in one line: enter on a 20-bar Donchian high/low break filtered by EMA200, exit on the opposite 10-bar extreme, hard stop at 2×ATR. 9 USDT-perp pairs, max 4 open, \$200 dry-run wallet. Backtest reference (2024-06→2026-06): +40.9%, PF 1.15, win 30.8%, max DD 25.9%, ~0.8 trades/day.

Files you touch:

```
ftbot/  
  config.dry.json          # bot config – holds YOUR local secrets (never commit them)  
  start-bot.bat / .ps1     # one-click: pull main + start the dry-run bot  
  watch-bot.bat / .ps1     # same, auto-redeploys on each main push touching ftbot/  
  strategies/  
    ClaudeBreakout.py      # LIVE strategy – do not edit mid-window  
    breakout_variants.py   # parameter neighbours (robustness, backtest only)  
    breakout_research.py   # research levers: Risk/Corr/Pyramid/Chandelier/Pro  
    breakout_hyperopt.py   # ClaudeBreakoutH0 – hyperopt-ready  
    research_sizing.py     # pure sizing math (unit-tested)  
    README.md             # strategies index  
    RESEARCH_returns.md    # the returns-optimization research  
    PC_GUIDE.md           # this file
```

2. Install freqtrade

Option A — Docker (recommended; Windows/Mac/Linux)

```
docker pull freqtradeorg/freqtrade:stable
```

Prefix every `freqtrade ...` command with (run from the repo root):

```
docker run --rm -v "$PWD/ftbot:/freqtrade/user_data" freqtradeorg/freqtrade:stable
```

For the long-running bot, drop `--rm` and add `-d --name ftbot --restart unless-stopped -p 127.0.0.1:8080:8080`.

With Docker, paths inside commands become `/freqtrade/user_data/...` instead of `ftbot/...`, and `--userdir /freqtrade/user_data`.

Option B — pip (Linux/WSL/macOS, Python 3.11+)

```
python -m venv ~/ft-env
source ~/ft-env/bin/activate      # Windows: ~\ft-env\Scripts\activate
pip install freqtrade            # pulls TA-Lib wheels on supported platforms
freqtrade --version
```

The commands below use Option B paths (`ftbot/...`); adjust for Docker as noted.

3. One-time setup

3a. Secrets in `config.dry.json`

The working-tree `config.dry.json` ships with placeholders. Fill in **locally** and **never commit** the real values (the repo keeps placeholders on purpose):

- `telegram.token` — a **new** bot token from @BotFather. Do **not** reuse `saif_trader_bot` (the Cloud Run relay holds a webhook on it; freqtrade polls; the two conflict — Telegram allows one consumer per token).
- `telegram.chat_id` — your existing chat id.
- `telegram.enabled` — set `true`.
- `api_server.jwt_secret_key`, `ws_token`, `password` — random strings.

3b. Download history

Live bot needs none, but backtests/hyperopt do. The strategy is 4h:

```
# enough history for the current backtest window
freqtrade download-data --userdir ftbot --config ftbot/config.dry.json \
  --timeframes 4h --timerange 20240101-

# for multi-regime confidence (2021 bull, 2022 bear) – see RESEARCH_returns §8
freqtrade download-data --userdir ftbot --config ftbot/config.dry.json \
  --timeframes 4h --timerange 20210101-
```

Futures funding data is pulled automatically for `trading_mode: futures`.

4. Run the live paper bot

Easiest — the batch scripts

```
ftbot\start-bot.bat      # pulls main, preserves your local secrets, starts dry-run
ftbot\watch-bot.bat     # same, and auto-redeploys whenever main changes ftbot/ (dry-run only)
```

Or directly

```
freqtrade trade --userdir ftbot --config ftbot/config.dry.json --strategy ClaudeBreakout
```

Keep it running 24/7: Docker `--restart unless-stopped`, or `tmux/systemd (pip)`. Optional dashboard: `freqtrade install-ui`, then `http://127.0.0.1:8080`.

You'll get a **breakout radar** Telegram push each 4h candle: per-pair bias and distance to the next Donchian trigger, plus open positions.

5. Monitor from your phone

Telegram commands (freqtrade native):

Command	Shows
<code>/status</code>	open trades + live P&L
<code>/profit</code>	cumulative profit summary
<code>/performance</code>	profit by pair
<code>/daily, /weekly, /monthly</code>	P&L by period
<code>/balance</code>	wallet
<code>/forceexit <id> / /forceexit all</code>	manual close (kill switch)
<code>/stopentry</code>	stop new entries, keep managing open ones

<code>/stop /start</code>	halt / resume the bot
<code>/count</code>	open vs max trades

Weekly review ritual (the main cadence)

Once a week, paste into a Claude cloud session for analysis:

```
# live dry-run trades
sqlite3 ftbot/tradesv3.dryrun.sqlite \
  "select pair, is_short, open_date, close_date, close_profit, exit_reason \
  from trades order by close_date desc limit 30;"
```

Or paste `/profit`, `/performance`, `/status` output. Compare against the backtest profile: win ~30%, winners avg ~+5% over ~6 days (Donchian exit), losers ~-4% (stop/trail). **Flag divergence; do not tune mid-window.** Losing streaks past 20 are within backtest norms — not a malfunction.

6. Research workflow (backtests on PC, interpret with Claude)

This is how you make the system better — all on the *research* variants, never the live bot. Full rationale: RESEARCH_returns.md. Do the steps **in order**.

Step 0 — Realism FIRST (decides if the edge is real)

Re-run the baseline with funding + a slippage assumption before building anything. A gross PF 1.15 can become net ~1.0 once you enter *into* momentum and pay funding on multi-day holds.

```
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json \
  --strategy ClaudeBreakout --timerange 20210101- --breakdown month
```

Add a slippage/fee assumption via config and confirm the net profit factor. If the honest $PF \leq 1.0$, stop and rethink — don't tune a non-edge.

Step 1 — A/B the levers against the live baseline

The research variants (built on `ClaudeBreakout`) each isolate one lever:

Strategy	Lever
<code>ClaudeBreakout</code>	baseline (live)
<code>ClaudeBreakoutRisk</code>	volatility-targeted unit sizing
<code>ClaudeBreakoutCorr</code>	+ correlation/crowding down-scaling
<code>ClaudeBreakoutPyramid</code>	+ Turtle pyramiding
<code>ClaudeBreakoutChandelier</code>	+ chandelier trailing exit

ClaudeBreakoutPro	all levers combined
-------------------	---------------------

```
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json \
  --strategy-list ClaudeBreakout ClaudeBreakoutRisk ClaudeBreakoutCorr \
    ClaudeBreakoutPyramid ClaudeBreakoutChandelier ClaudeBreakoutPro \
  --timerange 20240601- --breakdown month
```

Step 2 — Read the results the right way

Win rate is a red herring (this is a low-win, high-payoff system). Judge on:

- **Profit factor (PF)** — gross-profit / gross-loss, *net of fees+funding*.
- **Calmar** — return ÷ max drawdown. This is the primary metric (the go-live gate is a drawdown gate). Prefer a variant that lifts Calmar, not just raw return.
- **Max drawdown** — must stay within the gate ($\leq \sim 26\%$, ideally lower).
- **Expectancy / payoff ratio** — avg-win ÷ avg-loss; the levers should fatten this.
- **Per-month / per-pair breakdown** — confirm the edge isn't one pair or one quarter.

Step 3 — Walk-forward hyperopt (the "smarter" tuning)

ClaudeBreakoutH0 exposes entry/exit windows, trend EMA, ATR stop, and risk fraction as tunable parameters. **Optimise on history, validate untouched on a held-out period.**

```
# optimise on 2021-2024
freqtrade hyperopt --userdir ftbot --config ftbot/config.dry.json \
  --hyperopt-loss CalmarHyperOptLoss --strategy ClaudeBreakoutH0 \
  --timerange 20210101-20241231 --epochs 300 --spaces buy sell stoploss

# then VALIDATE the chosen params, unchanged, on 2025+
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json \
  --strategy ClaudeBreakoutH0 --timerange 20250101- --breakdown month
```

Adopt a parameter *region*, not the single best epoch — a real edge is a plateau, a spike is noise. The robustness table in `OVERHAUL_PLAN.md` already shows the slow gradient (30/15, EMA250) is the promising direction.

Step 4 — Decision rule

Keep a change only if it improves **Calmar on the validation period, net of funding+slippage**, without breaking the drawdown gate. One change per backtest so attribution stays clean. Bring the numbers to a Claude cloud session to sanity-check.

7. Optional deeper analysis

- **Monte-Carlo / trade-order shuffle** on the winning config to budget the 95th-pct drawdown — the single realised 25.9% DD is one sample; size against the distribution.
- **freqtrade backtesting-analysis** for entry/exit-reason and per-pair attribution.
- **freqtrade plot-dataframe / plot-profit** for visual sanity checks.

8. Go-live gates (ALL must pass before real money)

From `OVERHAUL_PLAN.md` §4 — these are the profit plan, do not weaken them:

- [] ≥ 4 weeks dry-run **and** ≥ 30 closed trades
- [] Dry-run profit factor ≥ 1.3 , win rate within ~ 10 pts of backtest
- [] Max dry-run drawdown $\leq 15\%$ of wallet
- [] No unexplained backtest-vs-dry-run divergence

Then: live Bybit keys, start with **\$50–100** (not full capital), **halve** position sizes for the first 2 weeks, and **tighten** the MaxDrawdown protection. Test one manual trade before enabling autonomous entries; watch the first 5 live trades closely.

9. Safety rules (read once, keep)

1. **Never** edit `ClaudeBreakout.py` or its parameters during the paper window.
2. **Never** commit your filled-in secrets in `config.dry.json` (Telegram token, jwt, password) — the repo keeps placeholders.
3. Research variants are **backtest-only** — none are wired into the live config.
4. Dry-run always **flatters**: real fills slip, funding drags, fees bite. The gates and the realism pass (§6 step 0) exist because skipping them is the fastest way to lose the real bankroll.
5. One hypothesis tweak per week max — no constant fiddling.

10. Troubleshooting

Symptom	Fix
Telegram silent	<code>telegram.enabled: true</code> , correct token/chat_id, bot /started, token not reused by the relay
"No data" in backtest	run <code>download-data</code> for 4h over the timerange first
Bot won't start after edit	<code>git stash local config</code> , <code>git pull</code> , re-apply secrets (this is what <code>start-bot.bat</code> automates)

Port 8080 in use	change <code>api_server.listen_port</code> or stop the other process
Hyperopt overfits	widen the validation window, prefer <code>CalmarHyperOptLoss</code> , adopt a region not a point
Strategy not found	check <code>--userdir ftbot</code> and the class name matches the file

11. Quick command reference

```
# download 4h data
freqtrade download-data --userdir ftbot --config ftbot/config.dry.json --timeframes 4h --timerange 20210101-20240601

# run live paper bot
freqtrade trade --userdir ftbot --config ftbot/config.dry.json --strategy ClaudeBreakout

# baseline backtest (with realism assumptions configured)
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json --strategy ClaudeBreakout --timerange 20210101-20240601

# lever A/B matrix
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json \
  --strategy-list ClaudeBreakout ClaudeBreakoutRisk ClaudeBreakoutCorr ClaudeBreakoutPyramid ClaudeBreakoutLeverage \
  --timerange 20240601- --breakdown month

# walk-forward hyperopt + validation
freqtrade hyperopt --userdir ftbot --config ftbot/config.dry.json --hyperopt-loss CalmarHyperOptLoss --timerange 20210101-20240601
freqtrade backtesting --userdir ftbot --config ftbot/config.dry.json --strategy ClaudeBreakoutH0 --timerange 20210101-20240601

# inspect live dry-run trades
sqlite3 ftbot/tradesv3.dryrun.sqlite "select pair,is_short,close_date,close_profit,exit_reason from trades"
```

Generated for the `trader-google-c / ftbot freqtrade` workstream. Pair this with `RESEARCH_returns.md` (rationale) and `strategies/README.md` (file index).